Results

For this Ben Olschar, 108021211678 and Frederik Hüttemann, 108021215247 cooperated with eachother. We did implement the code on our own, but agreed to using one version. The discussion and results are made in cooperation.

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Running Task 1: CPU Reduction
CPU-Avg-Time : 40.51799774ms

Running Task 2: GPU reduction global memory
Reductions match.

GPU-Avg-Time global memory: 20.93264818ms
Running Task 3: GPU cascaded
Before starting, we need to get some information about the GPU launch
configuration:
Number of SMs: 40
Max threads per SM: 1024
Max blocks per SM: 16
Thread warp size: 32
From this we can calculate an optimal launch configuration:
To maximize occupancy, we calculate BLOCK_SIZE as follows:
Suggested BLOCK_SIZE: 64
Suggested GRID_SIZE: 640
We also need to know the number of elements in our 40 MB float array:
A 40 MB array contains 10485760 float elements.

Reductions match.
GPU-Avg-Time global memory: 0.19919896ms
```

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GPU Configuration:

Parameter	**Value**
GPU	NVIDIA GPU Tesla T4
Compute Capability	7.5
Number of SMs	40
Max threads per SM	1024
Max blocks per SM	16
Warp size	32
Suggested BLOCK_SIZE	64

Parameter	Value
Suggested GRID_SIZE	640
Array size	40 MB
Number of float elements	10,485,760
Optimized BLOCK_SIZE (40 MB)	64
Optimized GRID_SIZE (40 MB)	640

Formatted results

Task	Description	Avg Time (ms)	Status
Task 1	CPU Reduction	40.517998	–
Task 2	GPU Reduction (global memory, atomic)	20.932648	Reductions match
Task 3	GPU Cascaded Reduction	0.199199	Reductions match